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Course: ITAI – 1371

Date: July 31, 2026

Lab 9 – Ensemble Methods

Dana Bolden: As part of this lab, I coordinated the project workflow and schedule management while also reviewing the execution scripts to make sure everything ran correctly. Before completing the assignment, I knew Decision Trees could be used for predictions, but I didn't fully understand why a Random Forest model performs better. Working through the notebook showed me how combining multiple Decision Trees creates more accurate and reliable predictions than relying on just one model. I also found it interesting to see the feature importance results, where petal length and petal width had the biggest impact on classifying the iris species. Seeing the code execute successfully alongside the visualizations helped connect the concepts to how the model makes decisions. Overall, this lab strengthened my understanding of ensemble methods and showed me how good planning, careful script review, and teamwork all contribute to a successful machine learning project.

Jonathan Aldana: I gained a practical understanding of ensemble methods and how combining multiple Decision Trees in a Random Forest creates a stronger, more reliable classifier. Running through the notebook allowed me to see how tree-based models calculate feature importances, revealing that petal length and petal width were the primary drivers for classifying iris species. After independently executing the code cells to verify that all the model outputs and visualization

plots ran smoothly, I analyzed the feature importance results to complete Knowledge Check Question 3. Overall, this lab reinforced how ensemble learning reduces model variance and helped me better interpret feature weightings in machine learning workflows

Astrid Lorenzana: I initially believed that only one model was needed for making predictions, that programmers, or anyone who worked on training the model, could only hone the model to become better at predicting and preventing the model from making errors. Learning that there are methods known as the “ensemble methods”, methods that either focus on an aspect of subsets in the data or models that correct errors made by a model that previously worked on the training data, this being very similar to working in real life as a team. The Random Forest technique is not just one model working on getting accurate predictions but rather many models. Upon training the Random Forest model, we were able to see a visual output under ‘Part 5: Feature Importance’ and because of that we are able to see a prediction that was made by putting together the predictions of other models together, allowing us to see a better made, more accurate prediction. Despite this lab not explicitly saying teamwork is a good way to get results, the message is there, and lots of people outside of our class could learn what teamwork is like based on the techniques learned here.